

EE115B *Digital Circuits*

Instructor: Chenxi Xiao

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Prof. Hengzhao Yang

Prof. Zhifeng Zhu

Prof. Yajun Ha



上海科技大学
ShanghaiTech University

1. SOP/Standard SOP
2. POS/Standard POS
3. Maxterm, Minterm.
4. Convert SOP to POS:
 - De Morgan theorem,
 - Truth table
 - K-Map
5. K-Map
 - SOP simplification
 - POS simplification

Boolean Algebra and Logic Simplification

- Boolean Algebra
- Logic Simplification
 - Karnaugh Map
 - 👉 Quine-McCluskey Method



Limitations of K-Map

- Karnaugh map is difficult to simplify more than 5 variables.
- Need something that can be readily programmed for computer.
- Quine-McCluskey Method: non-graphic, but can handle more variables.



Terminology

- Variable: the input
- Literal: a variable complemented or uncomplemented
- Implicant: a product term of literals obtained from the K-map such that when implicant = 1, $F = 1$
- Prime implicant (PI): The implicant which cannot be simplified further, i.e. no literal can be removed
- Essential prime implicant: The *prime implicants* which contain 1's that can only be grouped in 1 way (EPI)
- Cover: A set of *prime implicants* which covers all 1's

AB \ CD	00	01	11	10
00	1	1	0	0
01	1	1	1	0
11	0	0	1	1
10	1	1	0	0

- Variable: A, B, C and D
- Literal: A, B', etc
- Implicant: A'BC', AB'CD, A'D', etc
- Prime implicant (PI): A'C', ACD, ABD, BC'D, A'D', etc
- Essential prime implicant: ACD, not ABD, not BC'D, etc
- Cover: ACD, A'D', A'C' and (BC'D or ABD)

Quine-McCluskey Method

- Step 1: Write down function in SOP form

$$X = \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}CD + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D$$

- Step 2: Group them!
 - (Not though K-Map, but through a list)



Quine-McCluskey Method

- Step 2: Group them (convert to minterms, then group according to number of 1s)

$$XY + XY' = X$$

TABLE 4-9

<i>ABCD</i>	<i>X</i>	Minterm
0000	0	
0001	1	m_1
0010	0	
0011	1	m_3
0100	1	m_4
0101	1	m_5
0110	0	
0111	0	
1000	0	
1001	0	
1010	1	m_{10}
1011	0	
1100	1	m_{12}
1101	1	m_{13}
1110	0	
1111	1	m_{15}

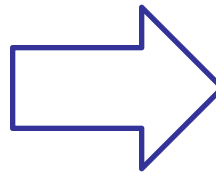


TABLE 4-10

Number of 1s	Minterm	<i>ABCD</i>
1	m_1	0001
	m_4	0100
2	m_3	0011
	m_5	0101
	m_{10}	1010
	m_{12}	1100
3	m_{13}	1101
4	m_{15}	1111

Quine-McCluskey Method

- Step 3: Reduce minterms by comparing adjacent groups.

TABLE 4-11

Number of 1s in Minterm	Minterm	ABCD	First Level
1	m_1	0001 ✓	(m_1, m_3) 00x1
	m_4	0100 ✓	(m_1, m_5) 0x01
2	m_3	0011 ✓	(m_4, m_5) 010x
	m_5	0101 ✓	(m_4, m_{12}) x100
	m_{10}	1010	(m_5, m_{13}) x101
	m_{12}	1100 ✓	(m_{12}, m_{13}) 110x
3	m_{13}	1101 ✓	(m_{13}, m_{15}) 11x1
4	m_{15}	1111 ✓	



Quine-McCluskey Method

- Step 3: Reduce minterms by comparing **adjacent** groups (2nd iteration).

TABLE 4-12

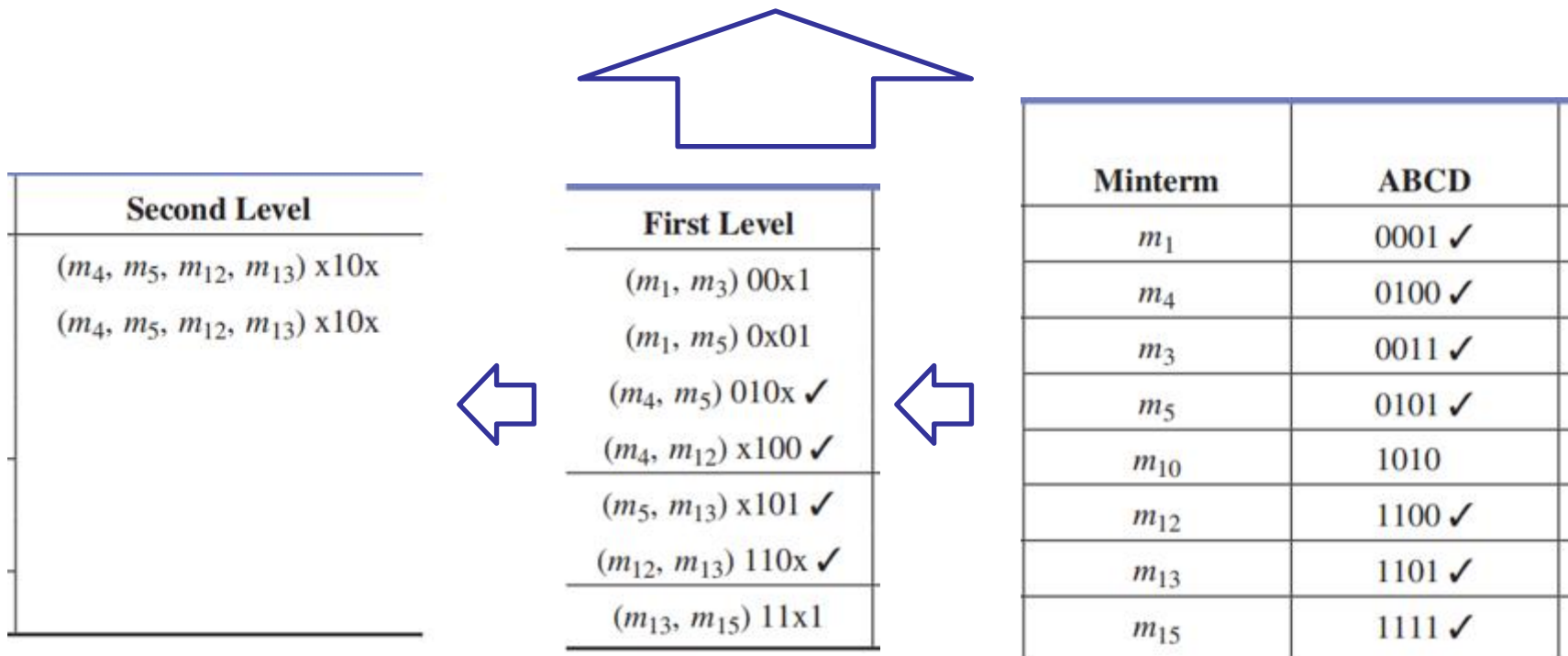
First Level	Number of 1s in First Level	Second Level
$(m_1, m_3) 00x1$ $(m_1, m_5) 0x01$ $(m_4, m_5) 010x \checkmark$ $(m_4, m_{12}) x100 \checkmark$	1	$(m_4, m_5, m_{12}, m_{13}) x10x$ $(m_4, m_5, m_{12}, m_{13}) x10x$
$(m_5, m_{13}) x101 \checkmark$ $(m_{12}, m_{13}) 110x \checkmark$	2	
$(m_{13}, m_{15}) 11x1$	3	



Quine-McCluskey Method

- Step 4: Write down reduced expression using **unchecked terms.**

$$X = B\bar{C} + \bar{A}\bar{B}D + \bar{A}\bar{C}D + ABD + \bar{A}\bar{B}C\bar{D}$$



Quine-McCluskey Method

- We have confirmed all PIs.
- But simply finding all PIs may not be the minimum possible expression (think why?).
- So we do an additional step for a final check.

Step 5: write prime implicant table (chart)

TABLE 4-13

Prime Implicants	Minterms							
	m_1	m_3	m_4	m_5	m_{10}	m_{12}	m_{13}	m_{15}
$B\bar{C} (m_4, m_5, m_{12}, m_{13})$			✓	✓		✓	✓	
$\bar{A}\bar{B}D (m_1, m_3)$	✓	✓						
$\bar{A}\bar{C}D (m_1, m_5)$	✓			✓				
$ABD (m_{13}, m_{15})$							✓	✓
$\bar{A}\bar{B}\bar{C}\bar{D} (m_{10})$					✓			

Quine-McCluskey Method

TABLE 4-13

Prime Implicants	Minterms							
	m_1	m_3	m_4	m_5	m_{10}	m_{12}	m_{13}	m_{15}
$B\bar{C} (m_4, m_5, m_{12}, m_{13})$			✓	✓		✓	✓	
$\bar{A}\bar{B}D (m_1, m_3)$	✓	✓						
$\bar{A}\bar{C}D (m_1, m_5)$	✓			✓				
$ABD (m_{13}, m_{15})$							✓	✓
$\bar{A}\bar{B}\bar{C}\bar{D} (m_{10})$					✓			

Before deciding which PI we need to remove (difficult decisions), it is simple to decide which PI we must keep (EPI).

If a minterm has a single check mark, then the prime implicant is essential and must be included in the final expression. The term ABD must be included because m_{15} is only covered by it. Likewise m_{10} is only covered by $\bar{A}\bar{B}\bar{C}\bar{D}$, so it must be in the final expression. Notice that the two minterms in $\bar{A}\bar{C}D$ are covered by the prime implicants in the first two rows, so this term is unnecessary. The final reduced expression is, therefore,

$$X = B\bar{C} + \bar{A}\bar{B}D + ABD + \bar{A}\bar{B}\bar{C}\bar{D}$$

Exercise

- Consider an example $f(a,b,c,d) = \sum m(0,1,2,5,6,7,8,9,10,14)$

→	0	0000	Group 0
→	1	0001	Group 1
	2	0010	
	8	1000	
	5	0101	Group 2
	6	0110	
	9	1001	
	10	1010	
	7	0111	Group 3
	14	1110	

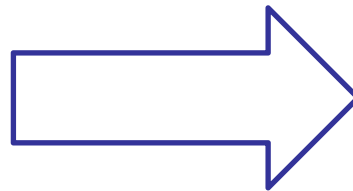
In the list, each term in group i has i 1's. E.g. all terms in group 2 have exactly two 1's

7



Exercise

Group 0	0	✓	0000
Group 1	1	✓	0001
	2	✓	0010
	8	✓	1000
Group 2	5	✓	0101
	6	✓	0110
	9	✓	1001
	10	✓	1010
Group 3	7	✓	0111
	14	✓	1110



0, 1	000-
0, 2	00-0
0, 8	-000
1, 5	0-01
1, 9	-001
2, 6	0-10
2, 10	-010
8, 9	100-
8, 10	10-0
5, 7	01-1
6, 7	011-
6, 14	-110
10, 14	1-10



Exercise

Group 0	0	✓	0000
Group 1	1	✓	0001
	2	✓	0010
	8	✓	1000
Group 2	5	✓	0101
	6	✓	0110
	9	✓	1001
	10	✓	1010
Group 3	7	✓	0111
	14	✓	1110

0, 1	✓	000-
0, 2	✓	00-0
0, 8	✓	-000
1, 5		0-01
1, 9	✓	-001
2, 6	✓	0-10
2, 10	✓	-010
8, 9	✓	100-
8, 10	✓	10-0
5, 7		01-1
6, 7		011-
6, 14	✓	-110
10, 14	✓	1-10

0, 1, 8, 9	-00-
0, 2, 8, 10	-0-0
0, 8, 1, 9	-00-
0, 8, 2, 10	-0-0
2, 6, 10, 14	--10
2, 10, 6, 14	--10



Exercise

Group 0	0	✓	0000	0, 1	✓	000-
	1	✓	0001	0, 2	✓	00-0
Group 1	2	✓	0010	0, 8	✓	-000
	8	✓	1000	1, 5		0-01
	5	✓	0101	1, 9	✓	-001
Group 2	6	✓	0110	2, 6	✓	0-10
	9	✓	1001	2, 10	✓	-010
	10	✓	1010	8, 9	✓	100-
	7	✓	0111	8, 10	✓	10-0
Group 3	14	✓	1110	5, 7		01-1
				6, 7		011-
				6, 14	✓	-110
				10, 14	✓	1-10

0, 1, 8, 9	-00-
0, 2, 8, 10	-0-0
2, 6, 10, 14	--10

Prime Implicants

1, 5	0-01	$a'c'd$
5, 7	01-1	$a'bd$
6, 7	011-	$a'bc$
0, 1, 8, 9	-00-	$b'c'$
0, 2, 8, 10	-0-0	$b'd'$
2, 6, 10, 14	--10	cd'



Prime Implicant Chart

Minterms	Prime Implicants	0	1	2	5	6	7	8	9	10	14
1, 5	$a'c'd$		X		X						
5, 7	$a'bd$				X		X				
6, 7	$a'bc$					X	X				
0, 1, 8, 9	$b'c'$	X	X					X	X		
0, 2, 8, 10	$c'd'$	X		X				X		X	
2, 6, 10, 14	cd'			X		X				X	X

- If a minterm is covered by only one implicant, the implicant is called **essential prime implicant**



Prime Implicant Chart

Iteration 1: Starting from EPI (solid line).

Iteration 2: m5 and m7 are not covered yet (dash line).

Find a way to include them:

Option 1: (1,5) and (6,7)

Option 2: (5,7)

Minterms	Prime Implicants	0	1	2	5	6	7	8	9	10	14	
1, 5	$a'c'd$		X		X							X
5, 7	$a'bd$				X		X					✓
6, 7	$a'bc$					X	X					X
0, 1, 8, 9	$b'c'$	X	X					X	X			✓
0, 2, 8, 10	$c'd'$	X		X				X		X		
2, 6, 10, 14	cd'			X		X				X	X	✓

■ Chosen implicants: $b'c'$ and cd'

■ Two minterms remaining: 5 and 7 – include $a'bd$

Minimal Cover

- In total, 3 prime implicants are needed to cover the function f

$$f = a'bd + b'c' + cd'$$

- Note that $a'bd$ is not an essential prime implicant
 - $m(5)$ and $m(7)$ are also covered in other PIs
 - $a'c'd$ and $a'bc$ could have also been chosen
 - But then, we won't have obtained minimum SOP



Alternatively, this can be derived using rules of boolean algebra

- Can be minimized by using consensus theorem

$$AB + A'C + BC = AB + A'C$$

$$f = a'c'd + a'bd + a'bc + b'c' + b'd' + cd'$$

$$f = a'c'd + a'bd + a'bc + b'c' + cd'$$

$$f = a'c'd + ((a'b)d + (a'b)c + cd') + b'c'$$

$$f = a'c'd + (a'b)d + cd' + b'c'$$

$$f = (a'd)c' + (a'd)b + b'c' + cd'$$

$$f = a'db + b'c' + cd'$$



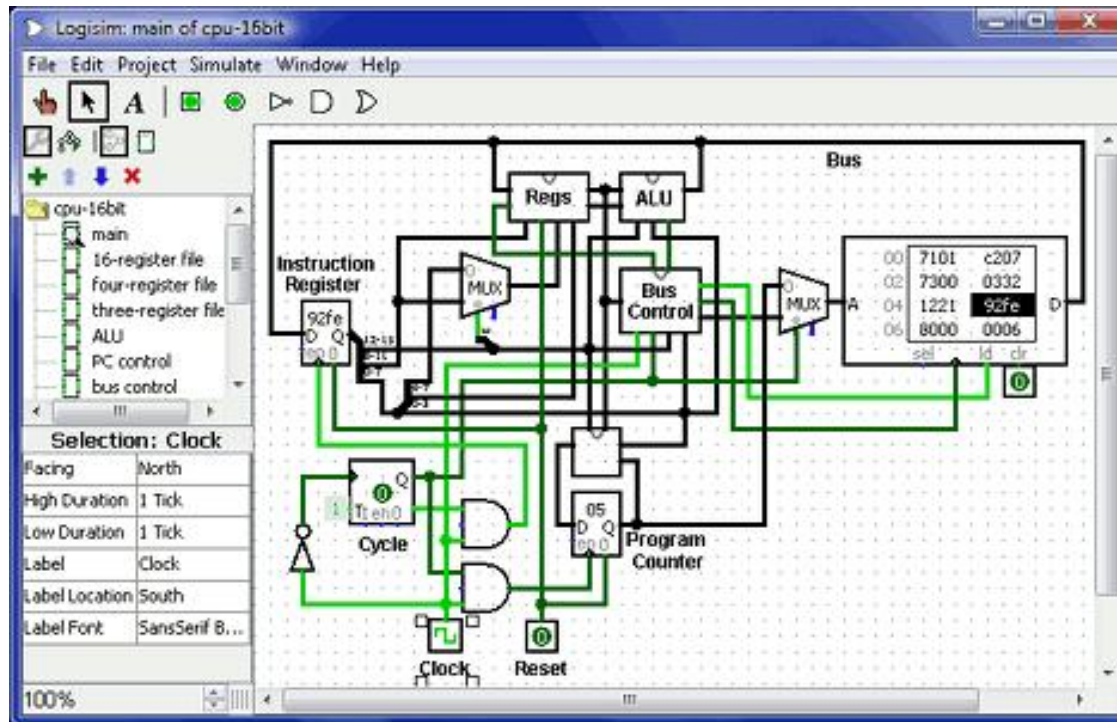
Reading Materials

1. <https://github.com/int-main/Quine-McCluskey>
2. Digital Fundamentals, Book 4-11, page237



Logisim Simulation

A free, open-source, graphical tool for logical simulation



<https://www.cburch.com/logisim/>

<https://github.com/logisim-evolution/logisim-evolution>



TI TINA Simulation

TINA-TI

SPICE-based analog simulation program

Downloads

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Overview

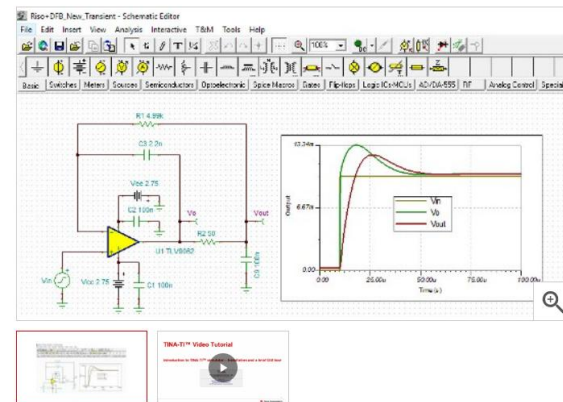
TINA-TI provides all the conventional DC, transient and frequency domain analysis of SPICE and much more. TINA has extensive post-processing capability that allows you to format results the way you want them. Virtual instruments allow you to select input waveforms and probe circuit nodes voltages and waveforms. TINA's schematic capture is truly intuitive - a real "quickstart."

TINA-TI installation requires approximately 500MB. Installation is straight-forward and it can be uninstalled easily, if you wish. We bet that you won't.

TINA is a product of DesignSoft exclusively for Texas Instruments. This complimentary version is fully functional but does not support some other features available with the full version of TINA.

For a complete list of available TINA-TI models, see: [SpiceRack – A Complete List](#)

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